SEMINAR REPORT

ON

Nanomaterial Application in Wastewater

By

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CERTIFICATE

This is to certify that Ms. Janhavi Modak (PRN: 1262242821)**,** a student of Third Year B.Tech. (Chemical Engineering), has successfully completed and presented the project seminar entitled "Nanomaterial Application in Wastewater" under my guidance and supervision during the academic year 2026–2027.

The seminar has been carried out in partial fulfilment of the requirements of the Project Seminar course prescribed in the curriculum of the Department of Chemical Engineering, Dr. Vishwanath Karad MIT World Peace University, Pune. The report has been prepared in accordance with the guidelines laid down by the Department, and the work presented is a bona fide record of the student's own efforts. The student has reviewed the relevant literature, analysed the subject in sufficient depth, and presented the findings satisfactorily before the review panel.

To the best of my knowledge and belief, the contents of this report are original, have been duly acknowledged wherever sourced from published literature, and have not been submitted elsewhere for any other academic purpose. The report has been checked for similarity and is within the permissible limits prescribed by the University.

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NOMENCLATURE

Symbol	Description	Unit
Roman symbols		
pH	Negative logarithm of hydrogen-ion activity	–
q_{max}	Maximum adsorption capacity	mg g^{-1}
T	Temperature	K
Abbreviations		
2D	Two-dimensional	–
As(V)	Arsenate (pentavalent arsenic)	–
BNNS	Boron nitride nanosheets	–
CBNM	Carbon-based nanomaterial	–
CNT	Carbon nanotube	–
CQD	Carbon quantum dot	–
g-C_3N_4	Graphitic carbon nitride	–
GO	Graphene oxide	–
h-BN	Hexagonal boron nitride	–
LDH	Layered double hydroxide	–
MB	Methylene blue	–
MO	Methyl orange	–
MOF	Metal–organic framework	–
NF	Nanofiltration	–
rGO	Reduced graphene oxide	–
RhB	Rhodamine B	–
RO	Reverse osmosis	–
RSM	Response surface methodology	–
UF	Ultrafiltration	–

ABSTRACT

Industrial growth, urbanisation and rising population have increased the volume of wastewater containing heavy metals, dyes, pharmaceuticals, pathogens and other persistent contaminants, many of which conventional physical, chemical, biological and membrane processes remove poorly. For chemical engineers, this is a separation and reaction-engineering problem with direct industrial consequences. This seminar reviews fifteen published studies on nanomaterial-based wastewater treatment. It classifies nanomaterials by dimensionality and composition and covers carbon-based materials such as graphene oxide (GO) and reduced graphene oxide (rGO), metal and metal-oxide nanoparticles, metal–organic frameworks, polymeric nanomaterials and hybrid systems. The mechanisms examined are adsorption, photocatalysis, membrane separation, antimicrobial action, redox reactions and sensing. The comparison shows that GO and its derivatives reach adsorption capacities of about 900 mg g^{-1} for Pb^{2+} and Cd^{2+}, and GO-containing membranes and composites achieve high removal of dyes, pharmaceuticals and metal ions. A magnetic GO–Gd_2O_3 nanocomposite reached 216.70 mg g^{-1} for As(V), with 99.9% maximum removal, and was tested on real mining-area water. Practical use, however, still depends on recovery, regeneration, environmental safety, cost and scale-up.

Keywords: *nanomaterials; wastewater treatment; adsorption; photocatalysis; graphene oxide; arsenic removal*

CHAPTER 1

Introduction

1.1 Background

Water is essential for human health, agriculture, industry and the ecosystems that people depend on. As industries expand, cities grow and populations rise, the volume of wastewater produced worldwide keeps increasing. When this wastewater is discharged without adequate treatment, it damages aquatic ecosystems and reduces the supply of clean water available for domestic and industrial use.

Wastewater often carries hazardous pollutants that do not break down easily, including heavy metals, dyes, pharmaceuticals, pathogens, pesticides and microplastics. Some of these are chemically very stable, some are present only at trace concentrations, and some resist standard treatment altogether. Physical separation, chemical treatment, biological processes and membrane filtration are all widely used, but they struggle with certain persistent and emerging contaminants (Aziz et al., 2025).

This gap has pushed researchers towards better treatment technologies, and nanotechnology has become one of the most active areas. At the nanoscale, materials behave differently from their bulk forms. Their large surface area relative to volume, strong surface reactivity, adjustable surface chemistry and catalytic ability all allow them to interact more effectively with pollutants (Salehpour et al., 2026). Nanomaterials can therefore treat wastewater in several ways, through adsorption, photocatalytic degradation, membrane separation, antimicrobial action and redox reactions. Carbon-based nanomaterials, metal and metal-oxide nanoparticles, metal–organic frameworks (MOFs), polymeric nanomaterials and nanocomposites have all been tested against a wide range of contaminants (Shrivastava et al., 2026).

Because one material can often perform several of these functions at once, nanomaterials make it possible to design treatment systems that are both more efficient and more flexible. There is a catch, however. Material cost, recovery, stability, possible environmental impacts and large-scale implementation all have to be resolved before such systems can be used in practice (Cheriyamundath and Vavilala, 2021).

1.2 Importance to Chemical Engineering and Industry

Wastewater treatment is, at its core, a chemical engineering problem. It combines separation processes such as adsorption and membrane filtration, reaction engineering such as photocatalytic and redox degradation, and the design of equipment that must operate continuously at scale. Nanomaterials change each of these. They raise the number of active sites available for adsorption, they act as catalysts or catalyst supports, and they alter the permeability and fouling behaviour of membranes (Salehpour et al., 2026; Bhuyan et al., 2026).

The industrial relevance is direct. Petrochemical effluents contain both organic and inorganic pollutants that must be removed together, and nanomaterial-based systems are being developed specifically for this combined load (Maikano et al., 2026). Pharmaceutical manufacturing, hospitals and aquaculture discharge antibiotics that conventional plants do not fully remove (Singh et al., 2021). Mining activity releases arsenic into ground and surface water, as in the field water tested by Lingamdinne et al. (2021). For each of these industries, the questions that decide adoption are the ones chemical engineers are trained to ask: how much material is needed, how it is recovered and regenerated, what it costs, and whether the process scales (Aziz et al., 2025).

1.3 Objectives

The objectives of this seminar are:

- to classify the nanomaterials used in wastewater treatment by dimensionality and composition, and to review the properties that make them effective;
- to explain the main treatment mechanisms, namely adsorption, photocatalysis, membrane separation, antimicrobial action, redox reactions and sensing;
- to compare the reported removal performance and limitations of selected nanomaterials;
- to examine two-dimensional carbon-based materials, including GO, rGO and related composites, in greater depth;
- to examine, through a case study, the removal of As(V) from contaminated water using a magnetic graphene oxide–gadolinium oxide ($GO–Gd_2O_3$) nanocomposite; and

- to identify gaps in recovery, regeneration, safety, cost and scalability, and to suggest directions for further work.

1.4 Scope of the Seminar

This report is a literature review. It does not present original experiments, simulations or laboratory measurements. Its sources are limited to the fifteen research and review papers listed in the References, and every numerical result reported here is taken from those papers and attributed to them.

The review covers nanomaterial classes, treatment mechanisms and application areas, including heavy metals, dyes, pharmaceuticals, pathogens and water-quality monitoring. It gives particular attention to 2D carbon-based nanomaterials because they offer high surface area and tunable surface chemistry, interact with pollutants in several ways, and can be combined with other materials to form hybrid nanocomposites. Throughout, each material is judged along the same path: its structure and properties, how it treats water, how well it removes pollutants, how stable it is and whether it can be regenerated, whether it can be recovered safely, and finally whether it is practical to use.

1.5 Organisation of the Report

Chapter 2 reviews the literature, beginning with conventional treatment and its limits, then the classes and properties of nanomaterials, and ending with the agreements, contradictions and gaps in the published work. Chapter 3 describes the review approach, the treatment mechanisms and the experimental methods reported in the cited studies. Chapter 4 presents and compares the reported results, including the GO–Gd_2O_3 case study, and discusses their practical implications. Chapter 5 summarises the conclusions, and Chapter 6 sets out the future scope of work.

CHAPTER 2

Literature Review

2.1 Conventional Wastewater Treatment and Its Limitations

Conventional treatment combines physical, chemical, biological and membrane processes. Each works well for some pollutants and poorly for others, as summarised in Table 2.1. Physical methods such as sedimentation, filtration and flotation are simple and cheap but leave dissolved contaminants untouched. Chemical treatment acts quickly on colour and toxic compounds but consumes chemicals and generates sludge. Biological processes are economical for biodegradable waste but fail with toxic and pharmaceutical contaminants. Membranes give excellent water quality but suffer from fouling and high energy demand (Aziz et al., 2025; Salehpour et al., 2026).

Table 2.1. Comparison of conventional wastewater treatment methods

Treatment method	Working principle	Advantages	Limitations
Physical (sedimentation, filtration, flotation)	Removes suspended solids by physical separation	Simple operation, low cost, suitable for large particles	Ineffective for dissolved contaminants, heavy metals and emerging pollutants
Chemical (coagulation, oxidation, precipitation)	Uses chemicals to destabilise or degrade pollutants	Rapid treatment; effective for colour and toxic compounds	High chemical consumption, sludge generation, secondary pollution
Biological (activated sludge, biofilm, anaerobic digestion)	Microorganisms biodegrade organic pollutants	Environmentally friendly; economical for biodegradable waste	Ineffective for toxic, non-biodegradable and pharmaceutical contaminants
Membrane (UF, NF, RO)	Separates contaminants by selective membrane filtration	High removal efficiency; excellent water quality	Membrane fouling, high energy demand, expensive maintenance

Source: Compiled from Aziz et al. (2025) and Salehpour et al. (2026).

The reviewed papers agree on the main point. Conventional methods handle bulk pollutants well, but they cannot fully remove persistent and emerging contaminants such as antibiotics, which pass through treatment plants and reach receiving waters (Singh et al., 2021). These shortcomings are what have driven the move towards nanomaterial-based treatment. Aziz et al. (2025) report removal efficiencies of 50–98% for organic contaminants and 80–99% for

inorganic pollutants across the nanomaterial systems they reviewed, which explains the interest. The same authors, however, list production cost, environmental risk and scalability as the barriers that keep these systems at small scale.

2.2 Classification and Properties of Nanomaterials

Nanomaterials are materials engineered with at least one dimension between 1 and 100 nm. Because so much of the material sits at the surface, they adsorb more, catalyse reactions better and react more readily than the same material in bulk (Salehpour et al., 2026). They are usually grouped by dimensionality into four classes, shown in Figure 2.1. Zero-dimensional (0D) materials are nanoscale in all three dimensions and include TiO_2, ZnO, Fe_3O_4, Ag and Au nanoparticles. One-dimensional (1D) materials, such as carbon nanotubes (CNTs), nanorods and nanowires, have a high aspect ratio that favours electron and mass transport. Two-dimensional (2D) materials are sheets or layers, such as graphene and GO, MXenes, hexagonal boron nitride (h-BN), layered double hydroxides (LDHs) and 2D-MOFs. Three-dimensional (3D) materials assemble nanoscale building blocks into bulk architectures such as nanocomposites, porous frameworks and aerogels.

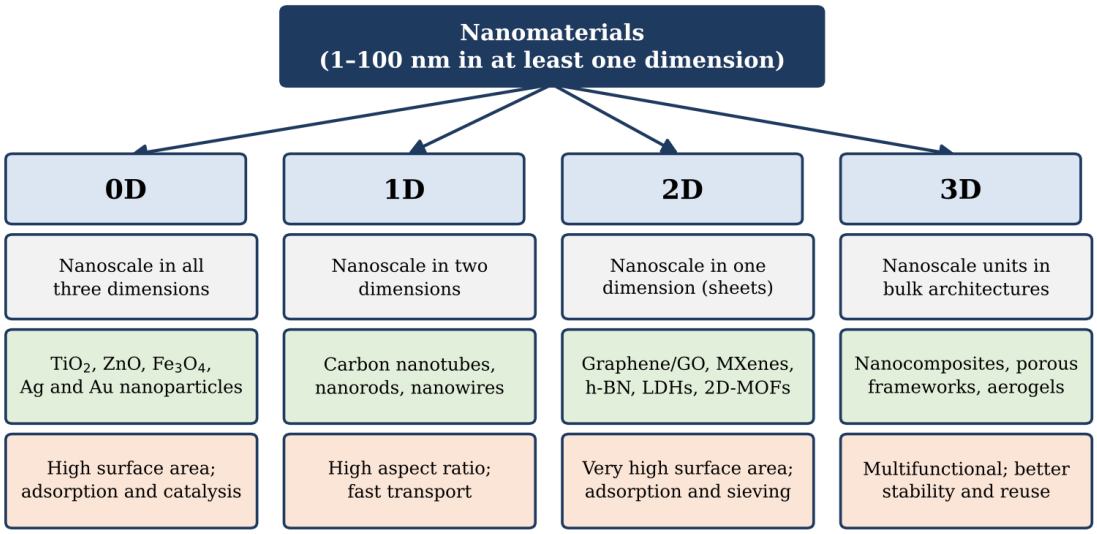


Figure 2.1. Classification of nanomaterials by dimensionality (0D, 1D, 2D and 3D), with examples and key features

Source: Original diagram prepared from the classification in Salehpour et al. (2026).

Dimensionality shapes surface structure, reactivity and transport behaviour, and these in turn decide how a nanomaterial performs in treatment. The properties that the reviewed papers

repeatedly link to performance are summarised in Table 2.2. Two points stand out. First, high surface area and tunable surface chemistry together explain why nanomaterials adsorb more and more selectively than conventional sorbents. Second, multifunctionality means that a single material can adsorb, degrade, filter, disinfect and sense, which is unusual among treatment media (Salehpour et al., 2026; Sheoran et al., 2022).

Table 2.2. Significance of nanomaterial properties in wastewater treatment

Property	Significance in wastewater treatment
High surface area-to-volume ratio	Provides more active sites for adsorption, raising pollutant removal efficiency
Enhanced catalytic activity	Accelerates photocatalytic degradation of organic contaminants under light irradiation
Tunable surface chemistry	Surface modification enables selective adsorption of specific pollutants
Size-dependent properties	Nanoscale dimensions influence optical, catalytic and physicochemical behaviour
High mechanical and chemical stability	Supports durability and reuse under different treatment conditions
Magnetic recoverability	Magnetic nanomaterials (e.g. Fe_3O_4) can be separated and reused after treatment
Multifunctionality	Allows simultaneous adsorption, degradation, antimicrobial action and sensing

Source: Compiled from Salehpour et al. (2026) and Sheoran et al. (2022).

2.3 Synthesis of Two-Dimensional Nanomaterials

Nanomaterials are made either top-down or bottom-up. Top-down routes break bulk material into nanoscale pieces, while bottom-up routes build nanostructures from atoms, molecules or clusters. For 2D materials, Bhuyan et al. (2026) describe top-down exfoliation of layered crystals by mechanical peeling or shear, by ball milling, as used for boron nitride nanosheets (BNNS) from h-BN, and by electrochemical exfoliation of graphene in aqueous $(NH_4)_2SO_4$. These routes are summarised in Figure 2.2. The synthesis route matters beyond the laboratory, because it sets the cost, defect density and scalability of the final material. Paradva et al. (2023) argue that green synthesis routes, which use plant extracts and other benign reagents, can reduce both cost and environmental burden.

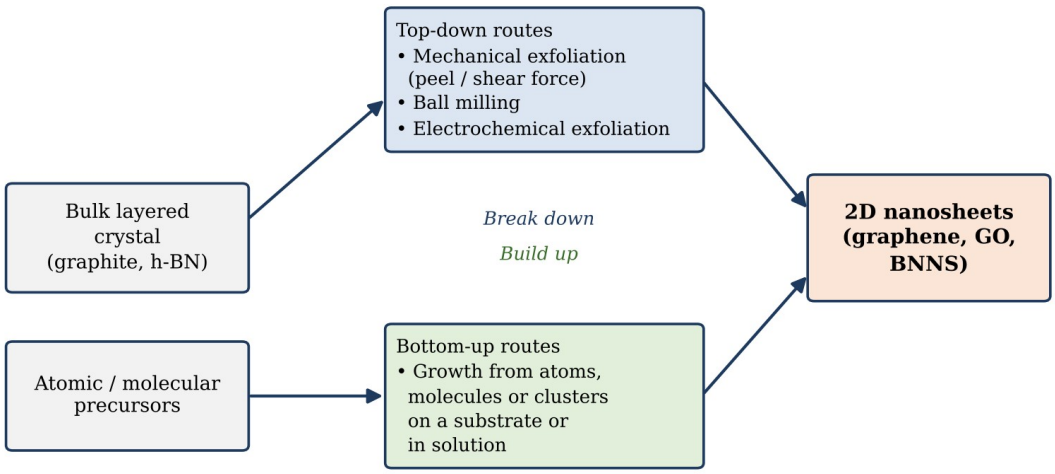


Figure 2.2. Top-down and bottom-up synthesis routes for 2D nanomaterials
Source: Original diagram prepared from the synthesis routes described in Bhuyan et al. (2026).

2.4 Carbon-Based Nanomaterials: GO, rGO and Related Materials

Carbon-based nanomaterials (CBNMs) include CNTs, graphene and GO, carbon nanospheres, fullerenes, carbon quantum dots (CQDs) and graphitic carbon nitride (g-C_3N_4) (Sheoran et al., 2022). Their large surface area, porosity, π-electron structure, thermal stability and tunable surface chemistry make them well suited to wastewater treatment. Sheoran et al. (2022) describe them as outstanding adsorbents for heavy metal ions, and Bhattacharyya et al. (2025) show that GO and rGO also remove dyes, pharmaceuticals and emerging pollutants. Nanomaterials, including carbon-based adsorbents, have also been reviewed for removing organic and inorganic pollutants together from petrochemical wastewater (Maikano et al., 2026).

Graphene oxide is the most studied member of this group. Its theoretical specific surface area is about 2600 m^2 g^{-1}, and its oxygen-containing groups bind contaminants through π–π interactions, electrostatic attraction and hydrogen bonding (Bhattacharyya et al., 2025). Functionalisation adds further surface groups that improve selectivity for specific pollutants. GO and rGO also serve as photocatalyst supports, membrane components and electrochemical materials, so a single family of materials can be used in many ways. When TiO_2 or ZnO nanoparticles are attached to graphene sheets, the graphene supports charge transport and complements the semiconductor, which can improve pollutant degradation (Thakre et al., 2021).

The main drawbacks reported for carbon nanomaterials are high production cost and possible environmental and ecotoxicological risks, which matter most at large scale (Sheoran et al., 2022; Bhattacharyya et al., 2025). Sheoran et al. (2022) also point to a lack of reliable tools for detecting these materials once released and of health-risk data on them.

2.5 Other Nanomaterial Classes and Hybrid Systems

Beyond carbon, the reviewed papers cover several other classes, compared in Table 2.3. Metal and metal-oxide nanomaterials such as TiO_2, ZnO and Fe_3O_4 are widely studied for photocatalysis, adsorption and pollutant removal, and Fe_3O_4 adds magnetic recoverability (Salehpour et al., 2026). MOFs offer high surface area and tunable porous structures for selective adsorption and separation. Polymeric nanomaterials provide functional groups and adjustable structures for adsorption and membrane treatment. Nanocomposites and hybrid systems combine materials so that the strengths of one offset the weaknesses of another (Shrivastava et al., 2026).

Table 2.3. Main nanomaterial classes, their mechanisms, targets and limitations

Nanomaterial class	Examples	Main mechanisms	Typical targets	Main limitation
Carbon-based	CNTs, GO, rGO, CQDs, g-C_3N_4	Adsorption; photocatalyst support; membranes	Heavy metals, dyes, pharmaceuticals	Production cost; ecotoxicity
Metal and metal oxide	TiO_2, ZnO, Fe_3O_4, Ag	Photocatalysis; adsorption; antimicrobial action	Dyes, organics, pathogens	Electron–hole recombination; recovery
Metal–organic frameworks	2D-MOFs and porous MOFs	Selective adsorption and separation	Selected pollutants	Not quantified in the reviewed papers
Polymeric	Functional polymer nanostructures, chitosan composites	Adsorption; membrane treatment	Metal ions, dyes	Not quantified in the reviewed papers
Nanocomposites and hybrids	TiO_2–GO, MoS_2–GO/chitosan, GO–Gd_2O_3	Combined adsorption, photocatalysis and separation	Mixed pollutant loads	Complexity; scale-up

Source: Compiled from Salehpour et al. (2026), Shrivastava et al. (2026), Sheoran et al. (2022), Bhuyan et al. (2026) and Lingamdinne et al. (2021).

Hybrid systems have attracted particular attention. Graphene–TiO_2 and graphene–ZnO composites improve photocatalytic treatment, but their performance depends on composite design, light conditions and electron–hole recombination, so laboratory efficiencies should not

be treated as plant-scale outcomes (Thakre et al., 2021). Photocatalytic nanohybrid membranes combine filtration with light-driven degradation and can offer self-cleaning and antifouling benefits (Kusworo et al., 2022). GO membranes, with their thin lamellar structure, hydrophilicity and tunable interlayer spacing, support selective transport, but the sheets swell in water and lose separation stability unless they are cross-linked (Pandey et al., 2022).

2.6 Agreements and Contradictions in the Literature

The reviewed studies agree on three points. All of them report that nanomaterials remove pollutants more effectively than conventional media under laboratory conditions (Aziz et al., 2025; Salehpour et al., 2026). All of them identify surface area and surface chemistry as the main reasons, whether the material is an adsorbent, a photocatalyst or a membrane additive (Sheoran et al., 2022; Bhattacharyya et al., 2025). And all of them conclude that cost, recovery and environmental safety, not removal performance, are what limit practical use (Cheriyamundath and Vavilala, 2021; Shrivastava et al., 2026).

The papers differ in where they place the main obstacle and how they propose to overcome it. Reviews of graphene adsorbents emphasise high capacity and good regeneration. Bhattacharyya et al. (2025) report that GO can be regenerated over more than six cycles with less than 10% loss in efficiency. Other authors stress that free nanosheets are hard to recover and propose immobilising them in 3D macrostructures (Abioye et al., 2024). Membrane studies are similarly divided. Pandey et al. (2022) see cross-linking as the route to stable GO membranes, while Kusworo et al. (2022) warn that photo-induced radicals can damage the polymer matrix of photocatalytic membranes and that catalysts can deactivate or release toxic metals. Photocatalysis itself is presented as a low-cost, efficient way to mineralise pharmaceuticals by Singh et al. (2021), but Thakre et al. (2021) caution that its performance depends strongly on light conditions and composite design.

A further difference concerns how results are reported. Most studies give the best removal or capacity achieved under optimised batch conditions, but they rarely use the same pollutant, concentration, pH or dose. Bhattacharyya et al. (2025) note that machine learning combined with response surface methodology (RSM) is now used to optimise adsorption conditions, which improves individual results but does not make different studies easier to compare.

Cheriyamundath and Vavilala (2021) observe that, despite rapid progress, information on practical implementation remains limited.

2.7 Research Gaps

The literature leaves several gaps open (Salehpour et al., 2026; Cheriyamundath and Vavilala, 2021):

1. *High removal does not imply practical applicability.* Many nanomaterials perform well under controlled laboratory conditions, but their performance in real, multi-contaminant wastewater is less explored.
2. *Recovery and reusability receive limited attention.* Recovery, regeneration and long-term performance are often not investigated, which raises concerns about secondary pollution and material loss (Abioye et al., 2024).
3. *Cost and scalability remain major barriers.* Complex synthesis, expensive precursors and energy- or solvent-intensive preparation limit large-scale use (Aziz et al., 2025).
4. *Environmental safety is under-reported.* Leaching, toxicity, aggregation and the long-term environmental fate of nanomaterials need more attention (Sheoran et al., 2022).
5. *Evaluation is fragmented.* Removal efficiency, stability, regeneration, recoverability and safety are rarely assessed together in the same study.

This seminar addresses these gaps by comparing the reported results along all of these criteria together, rather than by removal percentage alone, and by examining a case study in which recovery and real field water were part of the evaluation.

CHAPTER 3

Methodology

3.1 Review Approach

This seminar is a structured literature review. No experiments or simulations were carried out by the author. The review followed the six steps shown in Figure 3.1. First, recent research and review articles on nanomaterial-based wastewater treatment were collected; the fifteen papers listed in the References form the complete source base. Second, the nanomaterials were classified by structure, composition and treatment mechanism. Third, their reported removal performance, stability and limitations were compared. Fourth, 2D carbon-based materials, namely GO, rGO and related composites, were examined in more depth. Fifth, a single study was analysed as a case study. Sixth, research gaps and future directions were identified.

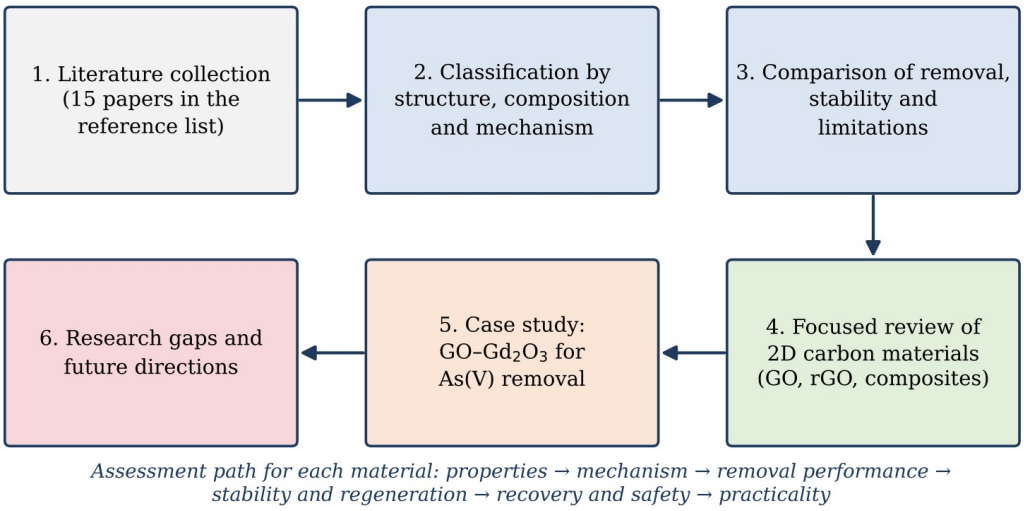


Figure 3.1. Flow diagram of the literature-review methodology

Source: Prepared by the author.

The work was divided into the six phases listed in Table 3.1. The sequence moves from literature to classification, then to mechanisms, performance comparison, limitations, research gaps and future directions.

Table 3.1. Phases of the literature review

Phase	Work carried out
Phase 1	Literature collection and screening
Phase 2	Classification of nanomaterials and treatment mechanisms
Phase 3	Comparative analysis of reported performance
Phase 4	Focused analysis of 2D carbon-based nanomaterials
Phase 5	Identification of research gaps and practical limitations
Phase 6	Review writing, synthesis and finalisation

Source: Prepared by the author.

3.2 Treatment Mechanisms Reported in the Literature

The reviewed papers describe six mechanisms by which nanomaterials remove or transform pollutants. In practice, several often act together (Salehpour et al., 2026; Cheriyamundath and Vavilala, 2021). They are summarised in Figure 3.2.

- *Adsorption.* Pollutants are captured on the nanomaterial surface through electrostatic interactions, hydrogen bonding, π–π interactions, ion exchange, pore filling or surface complexation. This is the dominant mechanism for GO-based adsorbents (Bhattacharyya et al., 2025).
- *Photocatalysis.* Light activates semiconductor nanomaterials such as TiO_2 or ZnO, generating reactive species that degrade persistent organic pollutants. Coupling with graphene can improve charge separation (Thakre et al., 2021).
- *Membrane separation.* Nanomaterials incorporated into membranes improve separation, permeability, selectivity and antifouling behaviour (Bhuyan et al., 2026; Pandey et al., 2022).
- *Antimicrobial action.* Reactive species and direct contact with microbial cells disrupt cellular structures and inhibit pathogens. The GO–Gd_2O_3 nanocomposite, for example, showed strong photocatalytic bacterial inactivation (Lingamdinne et al., 2021).
- *Redox reactions.* Reactive nanomaterials take part in electron-transfer reactions that transform or immobilise contaminants, particularly heavy metals (Salehpour et al., 2026).
- *Sensing.* Nanomaterial-based sensors provide sensitive and rapid detection of trace pollutants, supporting real-time water-quality monitoring (Salehpour et al., 2026).

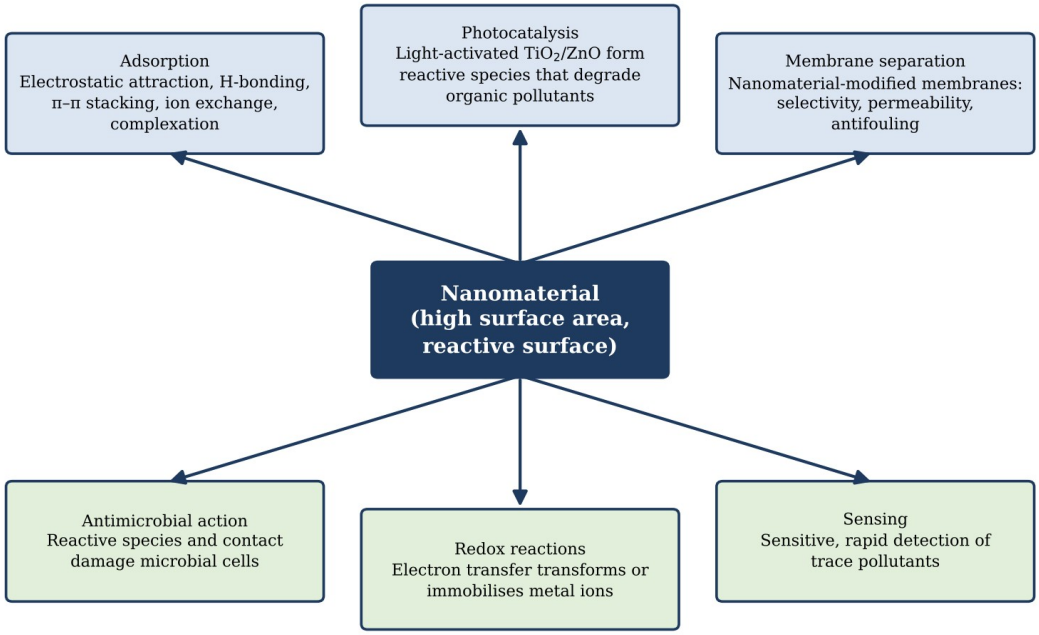


Figure 3.2. Mechanisms of nanomaterial-based wastewater treatment

Source: Original diagram prepared from the mechanisms described in Salehpour et al. (2026) and Cheriyamundath and Vavilala (2021).

3.3 Experimental Methods Reported in the Cited Studies

The cited studies rely mainly on laboratory batch experiments. In a typical adsorption study, a known dose of adsorbent is added to a solution of the target pollutant at a fixed pH and temperature, and the remaining concentration is measured. The results are expressed as adsorption capacity, in mg of pollutant per g of adsorbent, and as percentage removal. Lingamdinne et al. (2021) followed this approach for As(V), at an adsorbent dose of 0.1 g L^{-1}, pH 6.0 and 293 K, and then tested the material on real water from a mining area. Singh et al. (2021) review how such data are interpreted through adsorption kinetics and degradation pathways for pharmaceuticals. Bhattacharyya et al. (2025) describe the use of RSM, combined with machine learning, to optimise operating conditions.

Photocatalytic studies follow a similar pattern under controlled illumination, recording the degradation of a model dye or pharmaceutical over time (Thakre et al., 2021; Paradva et al., 2023). Membrane studies report rejection of model solutes, together with permeability and fouling behaviour, often for GO-based or nanohybrid membranes (Bhuyan et al., 2026; Kusworo et al., 2022; Pandey et al., 2022). Pilot-scale studies are rare. Bhattacharyya et al.

(2025) review some pilot-scale cases for graphene adsorbents, but most of the evidence in the reviewed literature comes from laboratory work.

3.4 Criteria Used for Comparison

To compare studies on a consistent basis, each material was assessed against the criteria in Table 3.2. These criteria follow directly from the research gaps identified in Section 2.7. They make it possible to judge a material not only by how much it removes but also by whether it can be recovered, reused and deployed safely.

Table 3.2. Criteria used to compare nanomaterial-based treatment systems

Criterion	What was examined	Reason for inclusion
Removal performance	Adsorption capacity (mg g^{-1}) or removal/rejection (%)	Primary measure of effectiveness
Mechanism	Adsorption, photocatalysis, membrane separation or combination	Explains selectivity and operating needs
Test conditions	Pollutant, dose, pH, temperature, synthetic or real water	Determines how far results can be compared
Stability and regeneration	Reuse cycles, loss of efficiency, swelling or deactivation	Governs operating cost and lifetime
Recovery	Magnetic separation, immobilisation, membrane form	Prevents secondary pollution and material loss
Safety, cost and scale	Leaching, toxicity, precursor cost, scale-up	Decides industrial feasibility

Source: Prepared by the author, based on the gaps identified by Salehpour et al. (2026) and Aziz et al. (2025).

CHAPTER 4

Results and Discussion

4.1 Conventional and Nanomaterial-Based Treatment Compared

Table 2.1 showed that each conventional method leaves a class of pollutants untreated: dissolved contaminants for physical treatment, sludge and secondary pollution for chemical treatment, toxic and pharmaceutical compounds for biological treatment, and fouling for membranes. Nanomaterial-based processes address several of these gaps. Aziz et al. (2025) report removal efficiencies of 50–98% for organic contaminants and 80–99% for inorganic pollutants across the systems they reviewed. The wide range for organics is itself informative: it reflects the variety of organic pollutants and the dependence of photocatalytic and adsorptive removal on operating conditions. The narrower and higher range for inorganic pollutants is consistent with the strong performance of carbon-based nanomaterials as heavy-metal adsorbents (Sheoran et al., 2022).

4.2 Adsorption Capacity of Graphene-Based Materials

Figure 4.1 compares the upper adsorption capacities reported by Bhattacharyya et al. (2025) for three classes of adsorbent. Activated carbon reaches about 260 mg g^{-1} of metal uptake and biochar about 600 mg g^{-1}, while GO and its derivatives reach about 900 mg g^{-1} for Pb^{2+} and Cd^{2+}. This is three to five times the capacity of conventional sorbents. The trend follows surface area and surface chemistry. GO combines a very high theoretical specific surface area, about 2600 m^2 g^{-1}, with oxygen-containing groups that bind metal ions directly (Bhattacharyya et al., 2025).

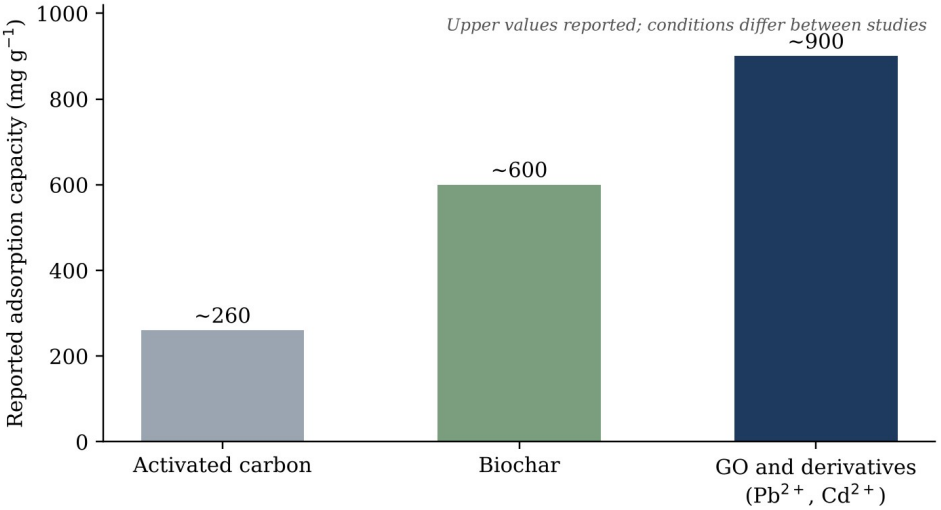


Figure 4.1. Upper adsorption capacities reported for activated carbon, biochar and GO-based adsorbents

Source: Plotted by the author from values reported in Bhattacharyya et al. (2025).

Two cautions apply. The values are upper limits taken from different studies, with different pollutants and conditions, so they indicate the potential of each material rather than its performance in a given plant. And a high capacity is only useful if the adsorbent can be regenerated. Here the evidence for GO is encouraging: Bhattacharyya et al. (2025) report regeneration over more than six cycles with less than 10% loss in efficiency.

4.3 Comparative Performance of Selected Nanomaterials

Table 4.1 sets out the reported performance of selected nanomaterials and nanomaterial-based systems, with their mechanism and main limitation. Each value comes from a different study, so the numbers should be read as indicative rather than strictly comparable.

Table 4.1. Comparative analysis of selected nanomaterials for wastewater treatment

Nanomaterial / system	Target pollutant	Mechanism	Reported performance	Key limitation	Source
GO-based membrane	Rhodamine B, Cd^{2+}	Adsorption	85.7 mg g^{-1} RhB; 60 mg g^{-1} Cd^{2+}	Stability and recovery	Bhuyan et al. (2026)
TiO_2–GO membrane	Dyes, pharmaceuticals	Photocatalysis + membrane separation	100% dye rejection; 95.7% pharmaceutical removal	Fabrication and scale-up	Bhuyan et al. (2026)
MoS_2–GO/ chitosan composite	MB, Pb^{2+}, Co^{2+}, Cu^{2+}, Ni^{2+}	Photocatalysis + adsorption	>99% MB and Co^{2+}; 89–96% Pb^{2+}, Cu^{2+}, Ni^{2+}	Complex composite system	Bhuyan et al. (2026)

Nanomaterial / system	Target pollutant	Mechanism	Reported performance	Key limitation	Source
Green ZnO nanoparticles	MO, MB, Congo red	Photocatalytic degradation	87% MO; 75% MB; >96% Congo red	Depends on synthesis and operating conditions	Paradva et al. (2023)
GO and derivatives	Pb^{2+}, Cd^{2+}	Adsorption	Up to ~900 mg g^{-1}	Cost, stability and regeneration	Bhattacharyya et al. (2025)
Magnetic GO–Gd_2O_3 nanocomposite	As(V)	Adsorption / surface complexation	216.70 mg g^{-1}; 99.9% maximum removal	Scale-up not yet demonstrated	Lingamdinne et al. (2021)

Source: Compiled from Bhuyan et al. (2026), Paradva et al. (2023), Bhattacharyya et al. (2025) and Lingamdinne et al. (2021).

Several trends emerge from Table 4.1. Plain GO-based materials work mainly by adsorption. Hybrid systems such as the TiO_2–GO membrane and the MoS_2–GO/chitosan composite go a step further by combining photocatalysis with membrane separation or adsorption, so that dyes, pharmaceuticals and metal ions can be treated together (Bhuyan et al., 2026). This combination explains their near-complete removal figures: the membrane or adsorbent retains what the photocatalyst does not degrade, and vice versa.

Green ZnO nanoparticles degrade dyes photocatalytically, but their performance varies from dye to dye, from 75% for methylene blue to more than 96% for Congo red, and depends on the synthesis route and operating conditions (Paradva et al., 2023). This spread is consistent with the caution of Thakre et al. (2021) that semiconductor photocatalysts are sensitive to light conditions and electron–hole recombination. GO and its derivatives give the highest adsorption capacity in the table (Bhattacharyya et al., 2025).

The limitations column is the most revealing part of the table. None of the reported limitations concerns removal performance. They are stability and recovery, fabrication and scale-up, composite complexity, cost and regeneration. In other words, the materials already remove pollutants well in the laboratory, and the open problems are practical ones.

4.4 Photocatalytic and Membrane-Based Systems

The adsorptive use of ZnO and TiO_2 nanoparticles is shown schematically in Figure 4.2. In this configuration, nanoparticles are dispersed in the wastewater, capture pollutants on their surface and must then be separated so that treated water can be discharged. The same nanoparticles act

as photocatalysts under suitable illumination, generating reactive species that degrade organic pollutants (Thakre et al., 2021).

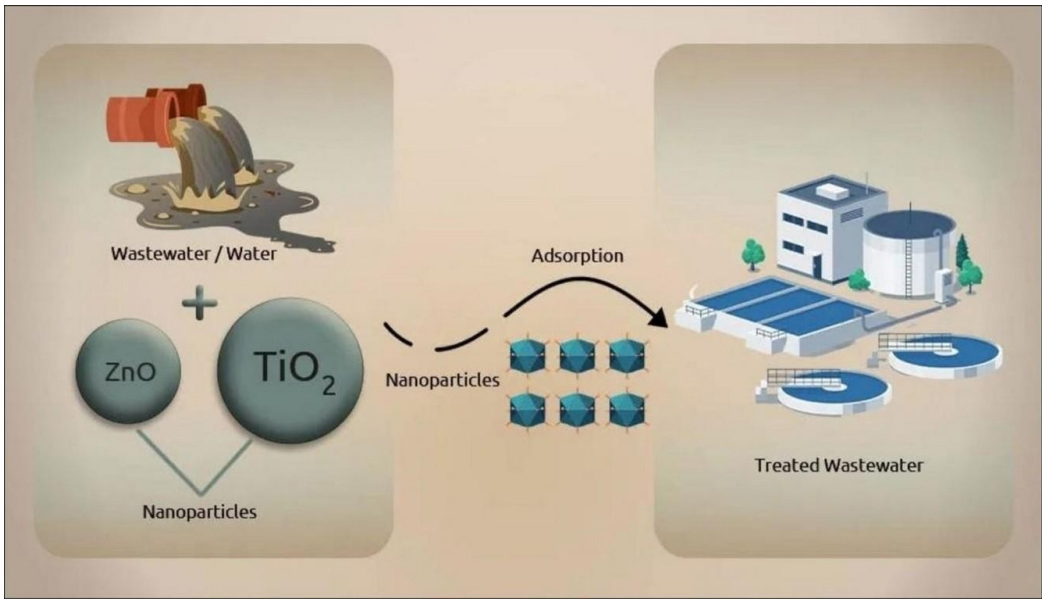


Figure 4.2. Application of ZnO and TiO_2 nanoparticles in wastewater treatment through adsorption, producing treated wastewater

Source: Prepared by the author for the seminar presentation.

The separation step is where dispersed nanoparticles become difficult, and it is the main reason for incorporating them into membranes. Photocatalytic nanohybrid membranes combine filtration with light-driven degradation and offer antifouling and self-cleaning benefits. Kusworo et al. (2022), however, identify catalyst deactivation, membrane fouling, possible release of toxic metals and damage to the polymer matrix by photo-induced radicals as unresolved issues. Material compatibility, illumination requirements, module design and operating cost therefore have to be considered together.

GO membranes face a different problem. Their lamellar structure and tunable interlayer spacing give selective transport, but the sheets swell in water, which widens the channels and reduces separation performance. Pandey et al. (2022) review chemical cross-linking of GO sheets with polymers as a way to improve hydration stability while preserving separation and antifouling behaviour. The trade-off is that cross-linked membranes still need long-term validation under realistic wastewater conditions.

4.5 Pharmaceuticals and Emerging Contaminants

Pharmaceutical residues, including antibiotics, persist through conventional treatment and enter receiving waters. Singh et al. (2021) identify hospitals, pharmaceutical plants, households, cattle farms and aquaculture as the main sources, and link antibiotic contamination to endocrine disruption and chronic toxicity. They review adsorption and photocatalysis as the main nanomaterial routes, with photocatalysis presented as a low-cost and efficient way to mineralise pharmaceutical compounds. They also argue that the nanomaterials can be recycled and the waste streams valorised, which places treatment within a circular economy. These approaches are promising, but performance should be assessed together with by-product formation, regeneration requirements and energy use.

Abioye et al. (2024) approach emerging contaminants from the point of view of point-of-use treatment, illustrated in Figure 4.3. Graphene nanosheets provide strong adsorption, but free nanosheets are difficult to recover. Assembling them into three-dimensional macrostructures improves handling and reuse and makes them suitable for filter- or cartridge-type devices. This illustrates a recurring design principle in the reviewed literature: high removal performance has to be paired with practical recovery and safe reuse.

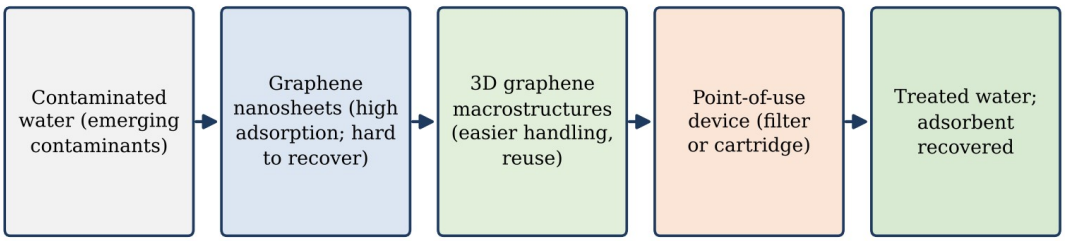


Figure 4.3. Point-of-use treatment pathway using graphene-based nanomaterials and macrostructures
Source: Original diagram prepared from the concept described in Abioye et al. (2024).

4.6 Case Study: GO–Gd_2O_3 Nanocomposite for As(V) Removal

Arsenic-contaminated water requires effective removal of As(V). Lingamdinne et al. (2021) prepared a magnetic hollow-sphere graphene oxide–gadolinium oxide (GO–Gd_2O_3) nanocomposite with a wool-like structure and a surface area of 50.91 $m^2\ g^{-1}$, and tested it for arsenic removal and antimicrobial activity. The conditions and results are listed in Table 4.2.

Table 4.2. Conditions and results for As(V) removal using the GO–Gd_2O_3 nanocomposite

Parameter	Value
Adsorbent	Magnetic hollow-sphere GO–Gd_2O_3 nanocomposite
Specific surface area	50.91 m^2 g^{-1}
Target pollutant	As(V)
Adsorbent dose	0.1 g L^{-1}
pH	6.0
Temperature, T	293 K
Maximum adsorption capacity, q_{max}	216.70 mg g^{-1}
Maximum As(V) removal	99.9%
Real-water test	Water from a mining area in South Korea; arsenic reduced below the maximum level permitted for drinking water
Additional function	Strong photocatalytic bacterial inactivation

Source: Lingamdinne et al. (2021).

Three features make this study a useful case. First, the capacity of 216.70 mg g^{-1} was achieved at a low dose of 0.1 g L^{-1} and near-neutral pH, and the authors report it as higher than or comparable to existing magnetic adsorbents (Lingamdinne et al., 2021). Second, the incorporation of Gd_2O_3 improved adsorption efficiency and increased the stability of the adsorbent compared with bare GO, while its magnetic character made the material easier to separate from solution. This directly addresses the recovery problem identified in Section 2.7. Third, the material was tested on real field water from a mining area, where it reduced arsenic below the drinking-water limit, rather than only on synthetic solutions.

As(V) is removed mainly by adsorption and surface complexation on the GO–Gd_2O_3 surface. The nanocomposite also showed antimicrobial activity through photocatalytic bacterial inactivation, so a single material addressed both a chemical and a biological contaminant (Lingamdinne et al., 2021). The study therefore illustrates the multifunctionality and magnetic recoverability described in Table 2.2. Its limits are also typical of the field: the work was carried out at laboratory scale, and behaviour at larger scale remains to be established.

4.7 Practical Implications: Recovery, Regeneration, Cost, Safety and Scale-Up

Taken together, the results in this chapter point to the same conclusion. The removal performance of nanomaterials is no longer the main obstacle. The barriers that remain are practical, and they are summarised in Figure 4.4 and Table 4.3.

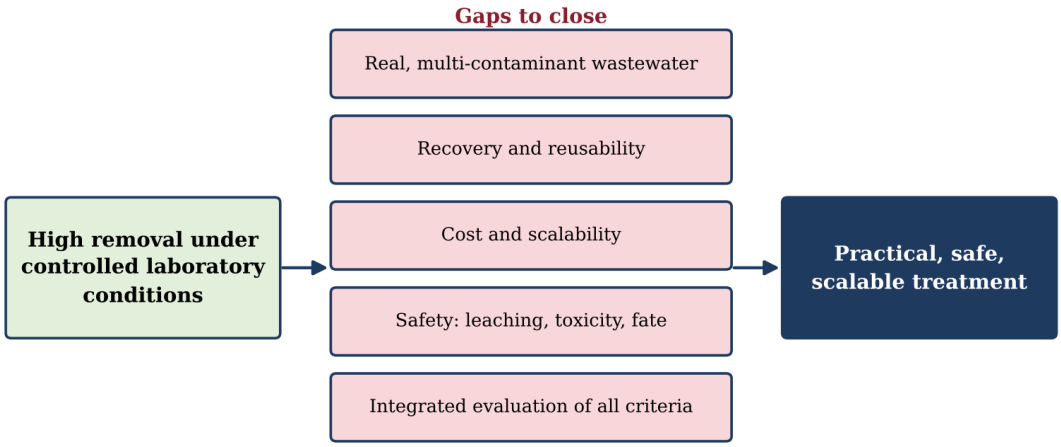


Figure 4.4. Gaps between laboratory performance and practical deployment of nanomaterial-based treatment

Source: Original diagram prepared from the gaps identified by Salehpour et al. (2026) and Aziz et al. (2025).

Table 4.3. Practical challenges and strategies reported in the reviewed literature

Challenge	Evidence from the literature	Strategy reported
Recovery	Free nanosheets and dispersed nanoparticles are hard to separate (Abioye et al., 2024)	Magnetic composites (Lingamdinne et al., 2021); 3D macrostructures (Abioye et al., 2024)
Regeneration	GO regenerated over >6 cycles with <10% efficiency loss (Bhattacharyya et al., 2025)	Report reuse cycles alongside capacity
Stability	GO membranes swell; photocatalysts deactivate (Pandey et al., 2022; Kusworo et al., 2022)	Cross-linking (Pandey et al., 2022); Gd_2O_3 incorporation (Lingamdinne et al., 2021)
Cost	High production cost of carbon nanomaterials (Sheoran et al., 2022; Aziz et al., 2025)	Green, low-cost synthesis (Paradva et al., 2023)
Environmental safety	Leaching, toxicity and fate under-reported (Sheoran et al., 2022)	Systematic risk assessment (Salehpour et al., 2026)
Scale-up	Most evidence comes from laboratory batch tests; some pilot-scale cases are reviewed by Bhattacharyya et al. (2025)	Validation on real wastewater and at pilot scale

Source: Compiled from the sources cited in the table.

Recovery is the most immediate issue, because nanoparticles that cannot be separated become a secondary pollutant. The reviewed papers offer two practical routes: magnetic composites such as GO–Gd_2O_3, and immobilisation of nanosheets in membranes or 3D macrostructures (Lingamdinne et al., 2021; Abioye et al., 2024). Regeneration determines operating cost, and the limited data available for GO are encouraging (Bhattacharyya et al., 2025). Stability

problems differ by system, from swelling in GO membranes to deactivation and matrix damage in photocatalytic membranes (Pandey et al., 2022; Kusworo et al., 2022).

Cost and environmental safety are linked. Expensive precursors and energy- or solvent-intensive synthesis raise the cost of carbon nanomaterials, and green synthesis routes are one proposed answer (Sheoran et al., 2022; Paradva et al., 2023). Environmental safety, including leaching, toxicity, aggregation and long-term fate, remains the least reported criterion in the reviewed studies (Salehpour et al., 2026). Finally, scale-up has barely begun. Most results in the reviewed papers come from laboratory batch tests, and only some pilot-scale cases have been reported (Bhattacharyya et al., 2025). Testing on real industrial effluents, such as petrochemical wastewater with its mixed organic and inorganic load (Maikano et al., 2026), is needed before nanomaterial systems can be designed into treatment plants.

CHAPTER 5

Conclusion

The main findings of this seminar are as follows:

1. Nanomaterials are a promising platform for advanced wastewater treatment because of their high surface area, enhanced reactivity and tunable surface chemistry.
2. Conventional physical, chemical, biological and membrane processes remain effective for bulk pollutants but are limited for dissolved, persistent and emerging contaminants; nanomaterial-based processes address several of these gaps (Table 2.1).
3. Nanomaterials act through adsorption, photocatalysis, membrane separation, antimicrobial action, redox reactions and sensing, and a single material can often combine several of these functions.
4. Dimensionality governs surface structure, reactivity and transport. 2D materials such as GO offer very high surface area and tunable chemistry, while 3D nanocomposites offer better stability and reuse.
5. GO and its derivatives reach adsorption capacities of about 900 mg g^{-1} for Pb^{2+} and Cd^{2+}, three to five times those of conventional sorbents, and GO-containing membranes and composites achieve high removal of dyes, pharmaceuticals and metal ions (Figure 4.1 and Table 4.1).
6. Hybrid systems that combine photocatalysis with adsorption or membrane separation give the most complete removal of mixed pollutants, at the cost of greater complexity.
7. The GO–Gd_2O_3 case study shows how high capacity can be combined with easier recovery: the nanocomposite reached 216.70 mg g^{-1} for As(V), with 99.9% maximum removal, and was tested on real field water (Lingamdinne et al., 2021).
8. The limitations reported across the literature are practical rather than performance-related: recovery, regeneration, stability, cost, environmental safety and scale-up. These must be evaluated together before nanomaterial-based treatment can be applied industrially.

CHAPTER 6

Future Scope of Work

The reviewed literature points to several directions for further work, each with direct industrial relevance.

- *Recovery and regeneration.* Future studies should report reuse cycles and efficiency loss alongside capacity, and should develop recoverable forms such as magnetic composites and 3D macrostructures (Bhattacharyya et al., 2025; Lingamdinne et al., 2021; Abioye et al., 2024). For industry, this determines adsorbent replacement cost and prevents secondary pollution.
- *Environmental safety.* Leaching, toxicity, aggregation and the long-term fate of nanomaterials need systematic assessment, together with tools to detect released materials (Sheoran et al., 2022; Salehpour et al., 2026). Without such data, regulatory approval for full-scale use is unlikely.
- *Sustainable, low-cost synthesis.* Green synthesis routes should be developed further to reduce precursor cost and environmental burden (Paradva et al., 2023).
- *Real-wastewater testing and pilot-scale validation.* Performance should be validated on real, multi-contaminant effluents and at pilot scale rather than only in laboratory batch studies (Bhattacharyya et al., 2025). Petrochemical, pharmaceutical and mining effluents are natural first targets (Maikano et al., 2026; Singh et al., 2021; Lingamdinne et al., 2021).
- *Stable membranes and hybrid systems.* Cross-linked GO membranes and photocatalytic nanohybrid membranes need long-term stability data under realistic conditions, and integrated systems combining nanomaterials with membranes, catalysts and sensors should be explored (Pandey et al., 2022; Kusworo et al., 2022; Bhuyan et al., 2026).
- *Process optimisation.* Statistical and data-driven tools such as RSM combined with machine learning can help optimise operating conditions and scale-up (Bhattacharyya et al., 2025).

- *Integrated evaluation.* Removal efficiency, stability, regeneration, recoverability, cost and safety should be evaluated together in the same study, so that materials can be compared on the criteria that decide industrial adoption (Salehpour et al., 2026; Aziz et al., 2025).

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